02/26/2026: Stochastic Calculus

CSCI 546: Diffusion Models

Textbook reference: Chang Sec 6.7-6.8, GS Sec 13.7

Announcement (Sign-in Sheet)

Please sign the sign-in sheet.

Review Problem Set #12

Concepts for Problem Set #13

Outline for today's material

- Interpreting a Stochastic Differential Equation (SDE)
- Difficulties with Integrating Against Brownian Motion

Stochastic Differential Equations

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This is known as a Stochastic Differential Equation (SDE). Our goal over the next two meetings is to make sense of SDEs.

Discretized SDEs

Definition. A SDE for the diffusion process X with drift coefficient $\mu(x, t)$ and diffusion coefficient $\sigma(x, t)$ is

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Poll. We've already seen one way to make sense of these. What is it?

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Solution.

$$\begin{cases} W \sim N(0,t) \\ \text{indep. increments} \end{cases} \implies \Delta W \sim N(0,\Delta t) = \sqrt{\Delta t}N(0,1)$$

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As $\Delta t \to 0$, this converges (under standard conditions) to the SDE

$$dX_t = \mu(X_t)\,dt + \sigma(X_t)\,dW_t$$

SDE: Shorthand for an integral equation

In general, we have

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$$\int dX(t) = \int \mu(X,t)dt + \int \sigma(X,t)dW(t)$$

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$$\int dX(t) = \underbrace{\int \mu(X,t)\,dt}_{\text{Riemann integral } \checkmark} + \underbrace{\int \sigma(X,t)\,dW(t)}_{???}$$

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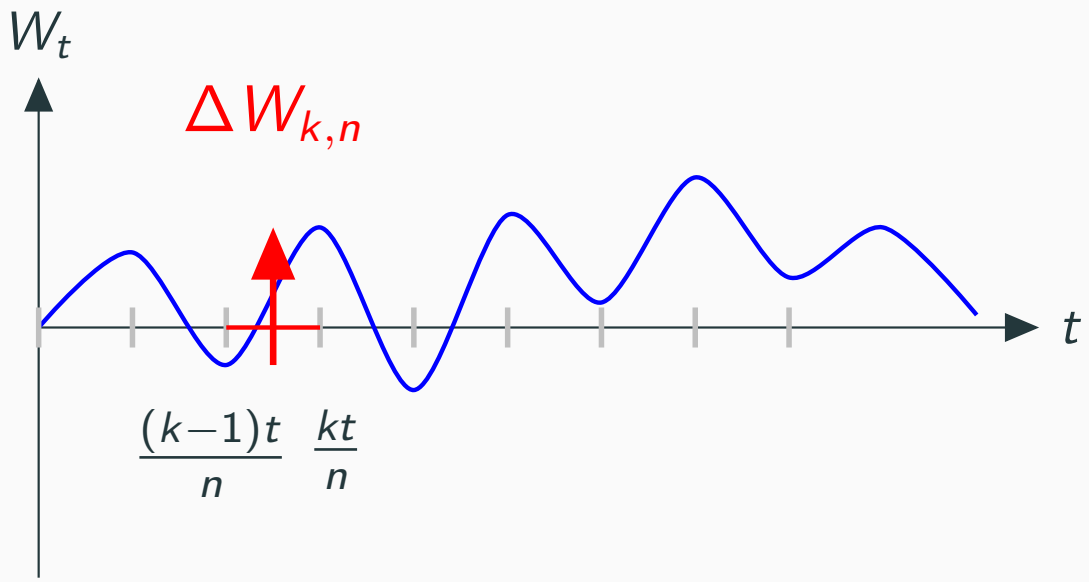
- Interpreting a Stochastic Differential Equation (SDE)
- Difficulties with Integrating Against Brownian Motion

"To alcohol! The cause of, and solution to, all of life's problems."

Similarly, the difficulty with (and solution to) integrating $\int f(t)\,dW(t)$ lies with the notion of the variation of a function.

Total variation of a function

First, form a **partition** of $[0, t]$ by chopping it into n bins.



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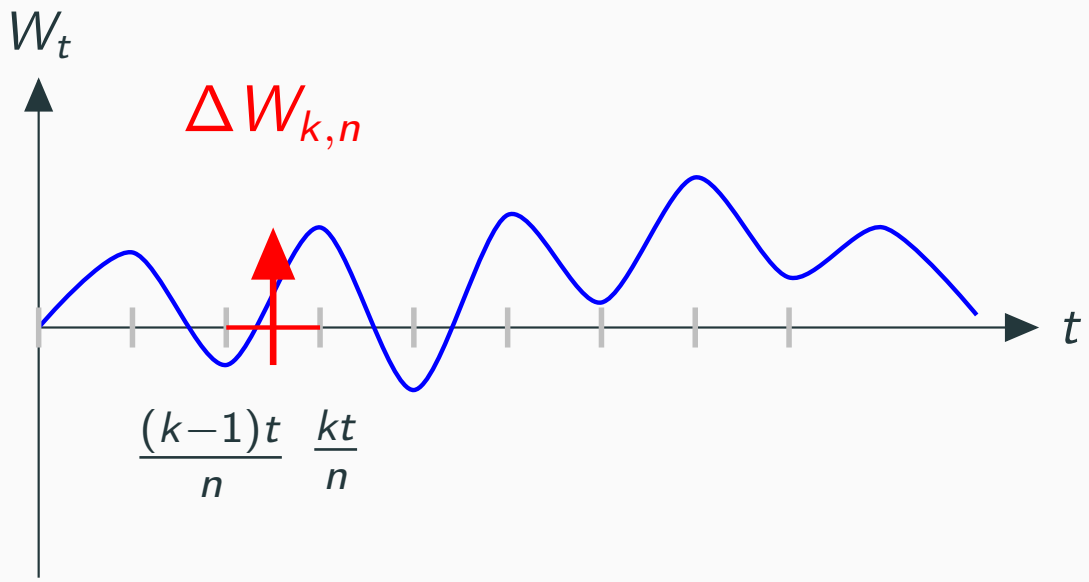

Total variation over a partition:

$$T_n = \sum_{k=1}^{n} |\Delta W_{k,n}|$$

Total variation: $\lim_{n\to\infty} T_n$

Brownian Motion has infinite total variation

Theorem

Let T be the total variation of standard Brownian motion over any interval $[0, t]$ where $t > 0$. Then $T = \infty$ with probability 1.

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Definition

The **Riemann integral** of a smooth function $f(t)$ on $[0, T]$ is given by

$$\int_0^T f(t)\,dt = \lim_{||P||\to 0} \sum_{i=1}^{n} f(t_i^*)(t_i - t_{i-1})$$

where $P = \{0 = t_0 < t_1 < \ldots < t_n = T\}$ is a partition of $[0, T]$, $||P|| = \max\{t_{i+1} - t_i\}$, and $t_{i-1} \leq t_i^* \leq t_i$.

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We'd like to use the above scheme, but for it to work directly, we need g to have **finite total variation**!

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This is the key to constructing the Itô integral.

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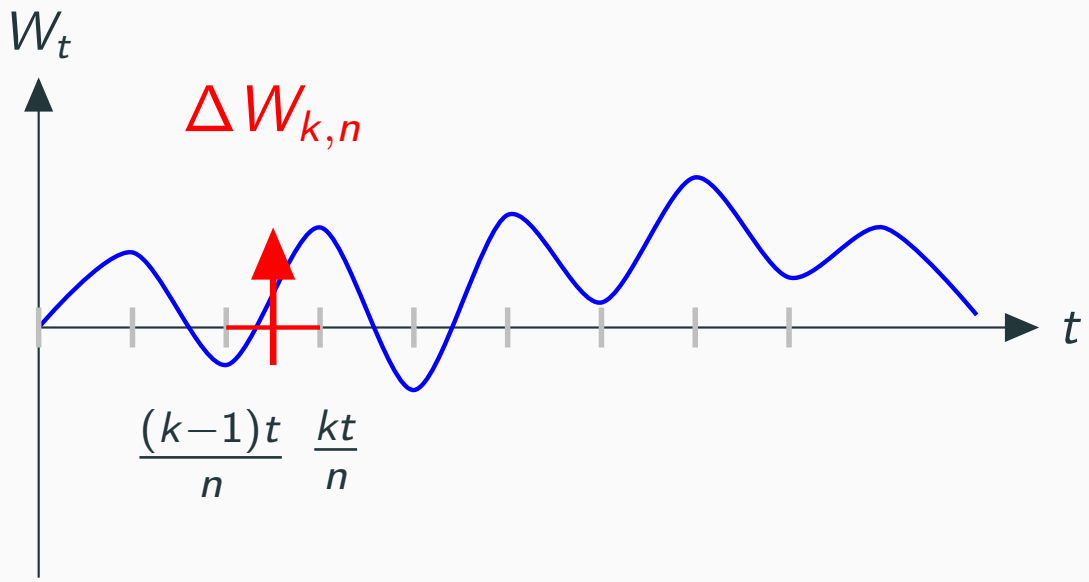

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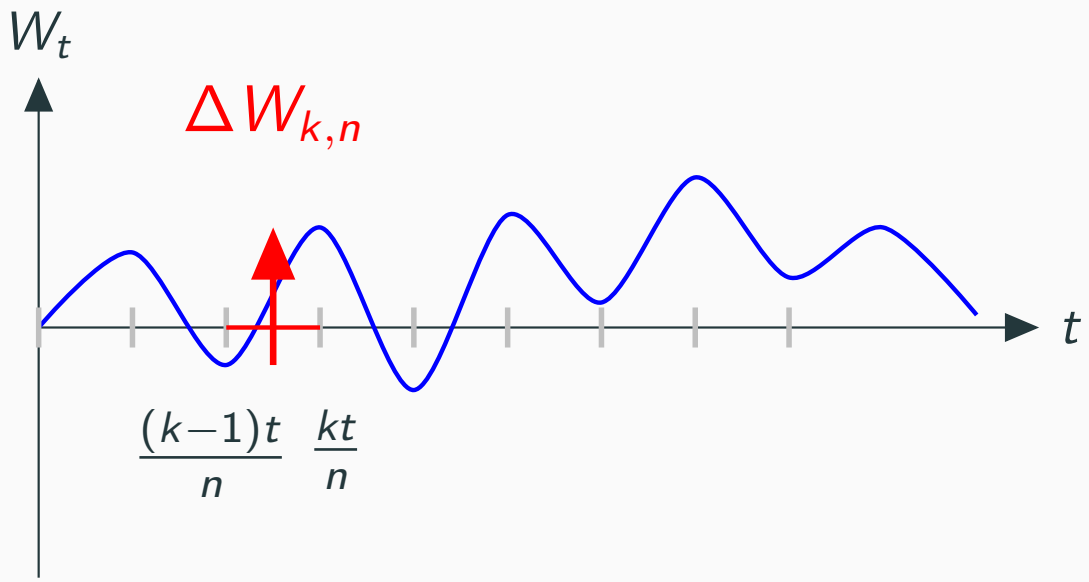

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Poll. Where does this strange relationship come from?

Let's Revisit: Brownian Motion as the continuous time limit of a Random Walk

Discrete model

$$X_n = \sum_{k=1}^{n} \xi_k, \qquad \xi_k = \begin{cases} +\Delta x & \text{w.p. } \frac{1}{2} \\ -\Delta x & \text{w.p. } \frac{1}{2} \end{cases}$$

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Poll. (Why does the above equation hold?)

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Requirement

To obtain a finite, non-trivial limit as $\Delta x, \Delta t \to 0$, we require

$$\boxed{\frac{(\Delta x)^2}{\Delta t} \to D > 0.}$$

Poll. Why do we want a finite, non-trivial limit?

Looking ahead

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To do so, we'll need a new tool called Itô Integration.

Random Groups

Aubrey Williams: 5
Austin Barton : 7
Blake Sigmundstad: 5
Diego Moylan: 6
Dillon Shaffer: 1
Ismoiljon Muzaffarov: 1
Jacob Tanner: 8
Josh Stoneback: 3
Joshua Bowen: 2
Joshua Culwell: 1
Laura Banaszewski: 2
Lina Hammel: 6
Logan Racz: 2
Matt Hall: 8
Mike Kadoshnikov: 3
Owen Cool: 4
Racquel Bowen: 4
Samuel Mocabee: 3
Tatiana Kirillova: 7

Group exercises - Problem Set 13

Theorem (Gautam Iyer)

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Theorem (Chang 6.7)

Let Q be the quadratic variation of standard Brownian motion over interval $[0, t]$. Then $Q = t$ with probability 1.

The exercises are related to the theorems above. For both exercises, let W be Brownian motion, and let $\Delta W_{k,n}$ denote the increment

$$\Delta W_{k,n} = W\left(\frac{kt}{n}\right) - W\left(\frac{(k-1)t}{n}\right)$$

over the k-th bin ($k = 1, \ldots, n$) of the uniform partition of $[0, t]$ with mesh size $\Delta t = \frac{t}{n}$.

1. (Gautam Iyer, *Stochastic Integration. (Lecture notes.)*) **Total Variation of Brownian Motion.** Let $T_n = \sum_{k=1}^{n} |\Delta W_{k,n}|$. Compute $\lim_{n\to\infty} \mathbb{E}[T_n]$.
2. (Chang Sec 6.7) **Quadratic variation of Brownian motion.** Let $Q_n = \sum_{k=1}^{n} (\Delta W_{k,n})^2$. Compute $\lim_{n\to\infty} \mathbb{E}[Q_n]$.